A Soft Introduction to Covariance Matrix Adaptation Evolution Strategy (CMA-ES)

From Gradient Descents to CMA-ES

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Introduction

The objective of the study is to understand the progression

gradient descent $\rightarrow$ random search $\rightarrow$ evolution strategy $\rightarrow$ CMA-ES

After: we shall see how important it is to adapt a full convariance matrix can make the algorithm nearly insensitive to rotations and anisotropic scalings of the search landscape.

Modified Objective: **Visit CMA-ES**

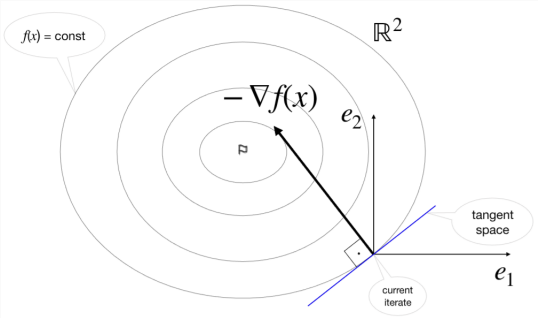
Optimization with Gradient Descents

Suppose we want to solve

$$\min_{x \in D} \quad f(x)$$

If f is smooth, we use gradient information

$$x_{n+1} = x_n - \sigma \nabla f(x_n)$$



Search information

GD uses gradient information to search for the optimizer.

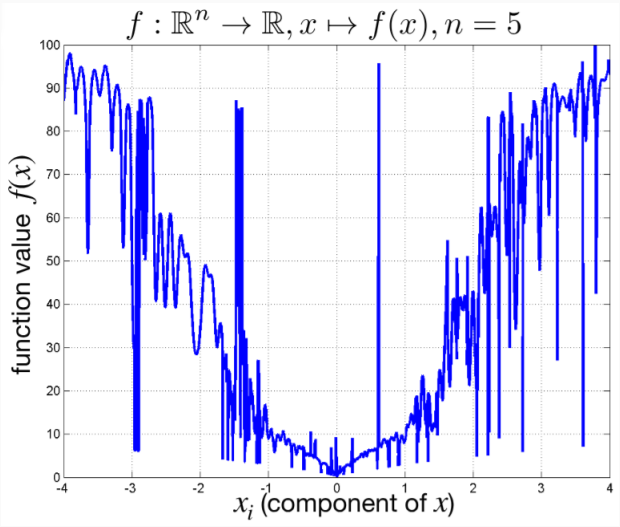


Figure 1: 5D Rugged Landscape [1]

[1] Hansen [2019]

Random Search

How can we reach the optimizer?

Random Search [2]

With a starting point $m \Rightarrow$ generate random points around $m \Rightarrow$ calculate the function values f.

Search information

The random search uses a point m to generate random points based on probability distribution and calculate the fitness value (function value).

Drawback: No directional learning/adaptation.

This is as stated " Finding a needle in an n-dimensional haystack " [3].

[2] White [1970]
[3] Korn [1966]

Evolution Strategy: (μ, λ)-ES

A simulation is here! [click me]

Search Information

(μ, λ)-ES uses a mean m generate to λ random points (called *offsprings*) and selects μ points (called *parents*), recombine to generate a new mean, and repeat the process.

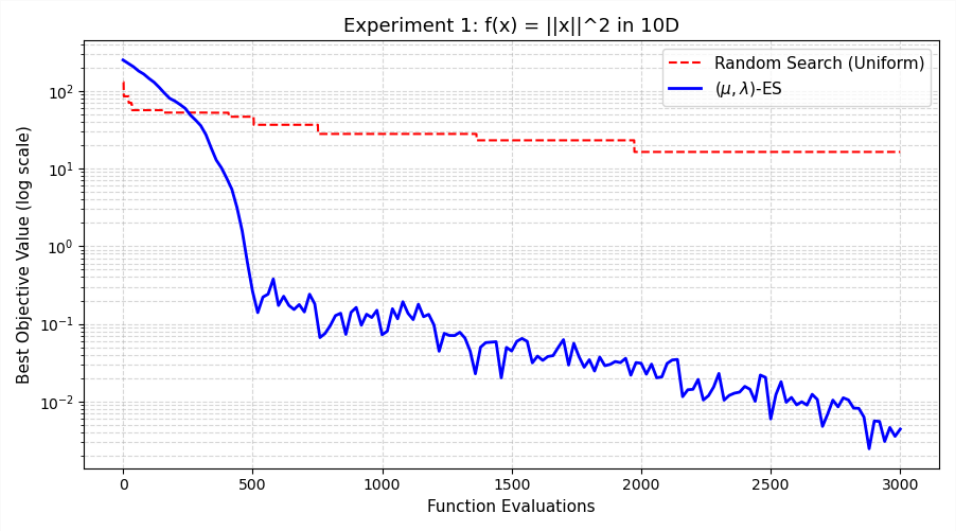


Figure 2: Comparison between random search with uniform distribution and (μ, λ)-ES.

Ill-Conditioned Problems

(μ, λ)-ES is not enough...

- One observation is that the algorithm is inefficient when the successful search directions from a thin elongated ellipsoid.

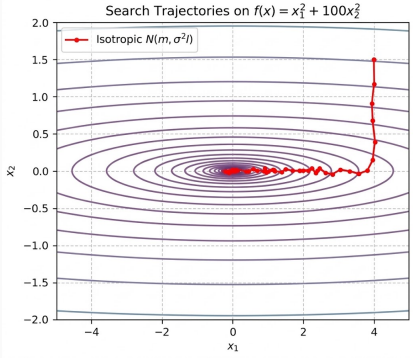
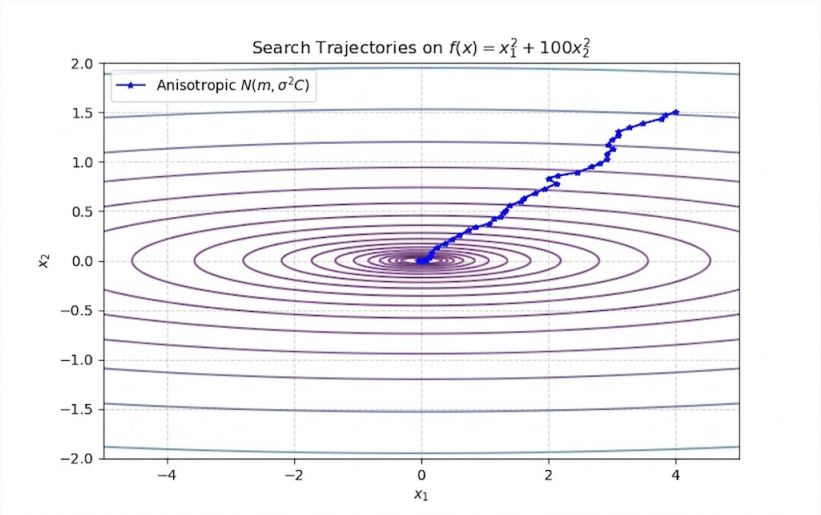


Figure 3: Function f has anisotropic level set. (Steep valey in x_2 and flat in x_1).

Preferable search trajectory



Drawback:

Have a direction, but ineffective with ill-conditioned problems.

Covariance Matrix Adaptation Evolution Strategy (CMA-ES)

Search Information (Expected)

CMA-ES $=$ (μ, λ)-ES $+$ learning/adaptation

(μ, λ)-ES generates points

$$X \sim \mathcal{N}(m, \sigma^2 I).$$

CMA-ES generates points

$$X \sim \mathcal{N}(m, \sigma^2 C)$$

where C is symmetric positive definite.

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CMA-ES generates points

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where C is symmetric positive definite.

What does the covariance matrix C do here?

A simulation is here! [click me]

- Directions in which successful steps have repeatedly occurred should receive larger variance in the future generations.

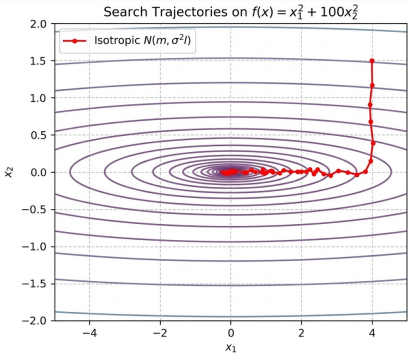


Figure 4: Function f has anisotropic level set. (Steep valley in x_2 and flat in x_1).

Search Information

CMA-ES **learns** not only where to search thought the starting point m but also how to search through C.

What information that C gives?[4]

$$C_{n+1} = (1 - c_\mu) C_n + c_\mu \frac{1}{\sigma^{(i)^2}} C^\mu_{n+1}$$

[4] Aguirre et al. [2026]

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[4] Aguirre et al. [2026]

There's something more...

We can control the step size σ[4]

$$\ln \sigma_{n+1} = \ln \sigma_n + \frac{c_\sigma}{d_\sigma} \left(\frac{\left\| p_{n+1}^\sigma \right\|}{E \left\| \mathcal{N}(0, I) \right\|} - 1 \right)$$

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The End of CMA-ES!!!

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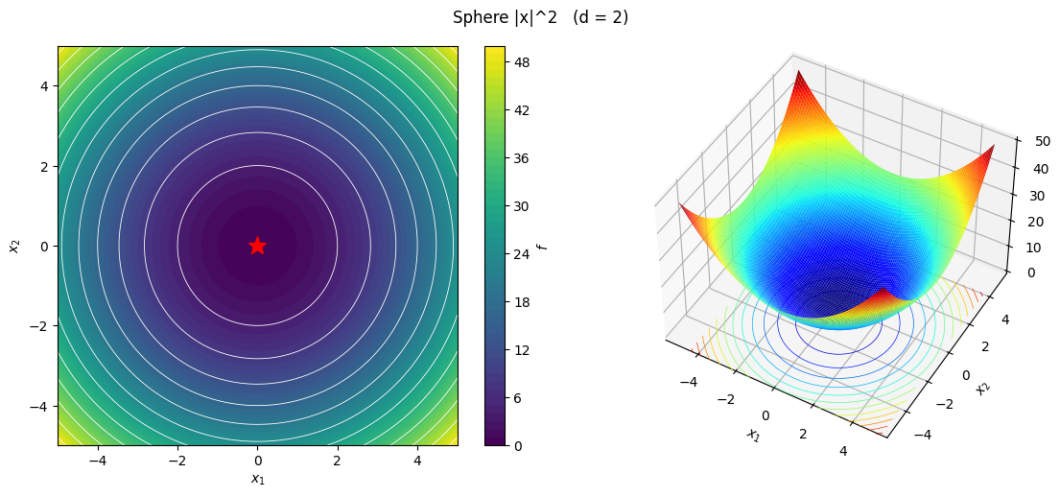
... from me

CMA-ES vs Gradient Descent vs Quasi-Newton

Five landscapes with known minimizer $\mathbf{x}^*$, tested in $d = 2$ and $d = 10$:

1. **Sphere** $|\mathbf{x}|^2$: isotropic, convex
2. **Ellipsoid** $\sum_i 10^{6i/(d-1)} x_i^2$: separable, condition number 10^6.
3. **Rotated ellipsoid** $\mathbf{x}^T Q^T \Lambda Q \mathbf{x}$: same spectrum, principal axes no longer aligned with the coordinates.
4. **Rosenbrock**: non-convex but *smooth* and unimodal; a narrow valley whose curvature changes along its length.
5. **Rastrigin**: multimodal, local minima on a unit grid; tests global behaviour rather than local convergence.

Isotropic Quadratic Problem



Convergence Analysis: $\|\mathbf{x}_n - \mathbf{x}^*\|$

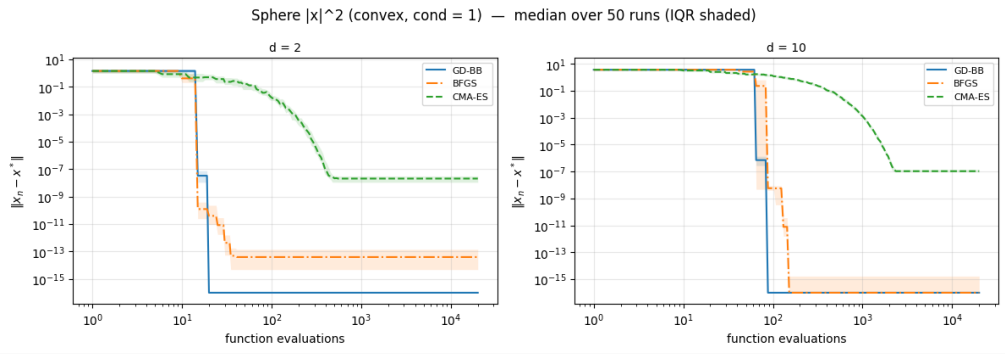
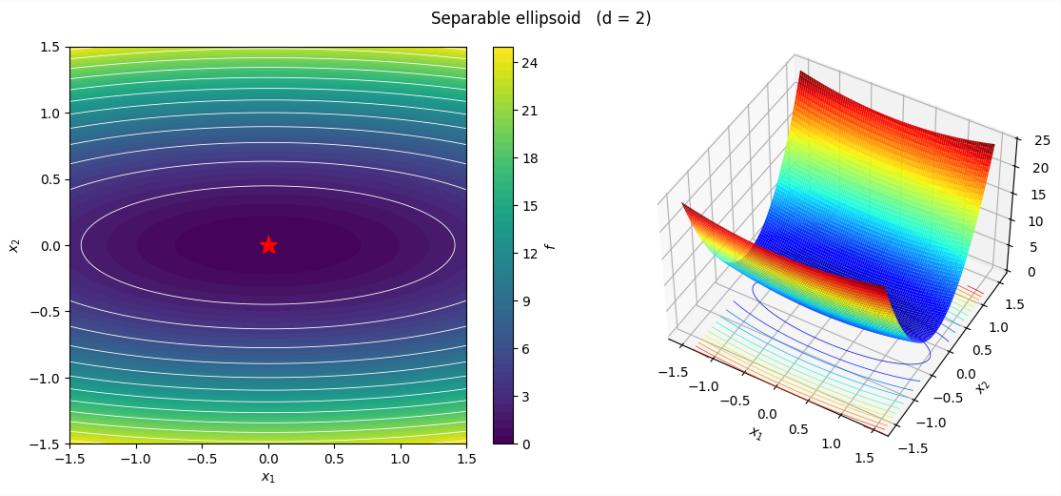


Figure 5: Convergence Analysis on Isotropic Quadratic Problem

Ill-Conditioned Quadratic Problem



Convergence Analysis: $\|\mathbf{x}_n - \mathbf{x}^*\|$

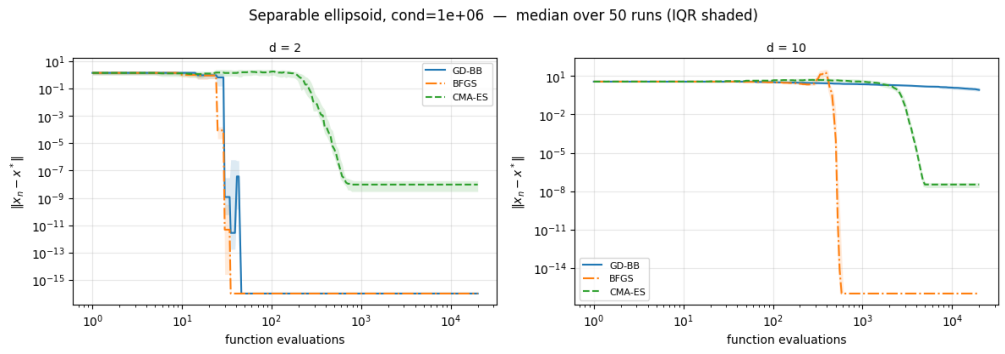
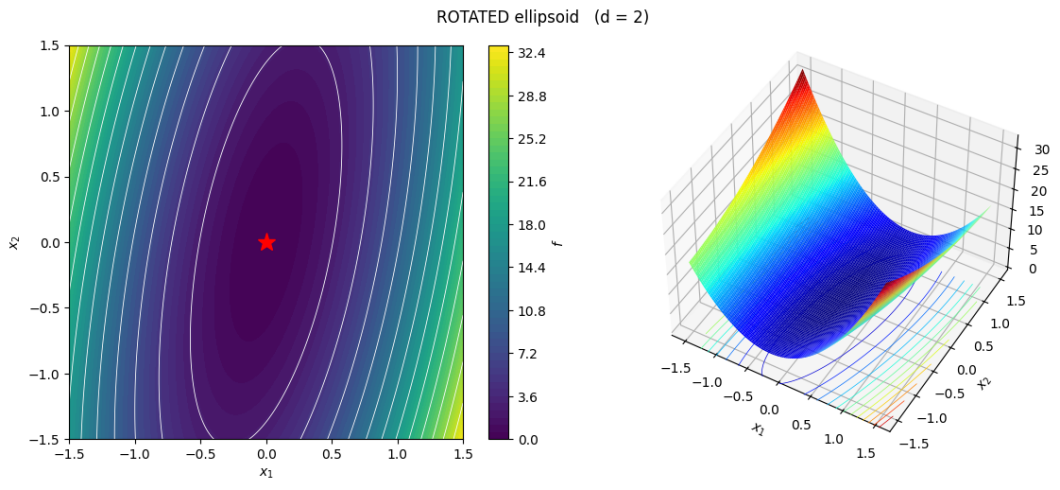


Figure 6: Convergence Analysis on Ill-Conditioned Quadratic Problem

Rotated Ellipsoid Problem



Convergence Analysis: $\|\mathbf{x}_n - \mathbf{x}^*\|$

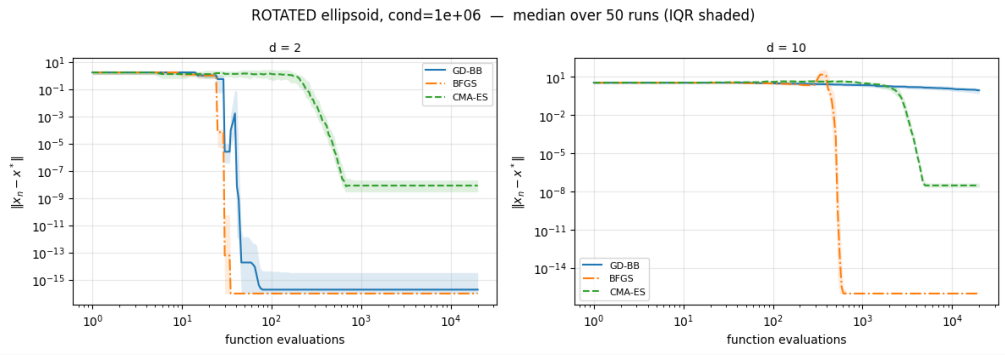
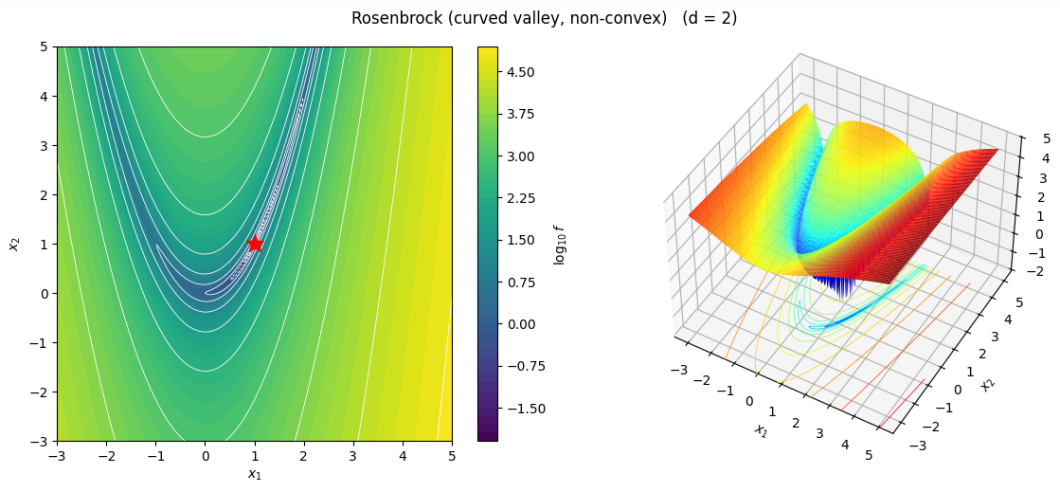


Figure 7: Convergence Analysis on Rotated Quadratic Problem

Rosenbrock Problem



Convergence Analysis: $\|\mathbf{x}_n - \mathbf{x}^*\|$

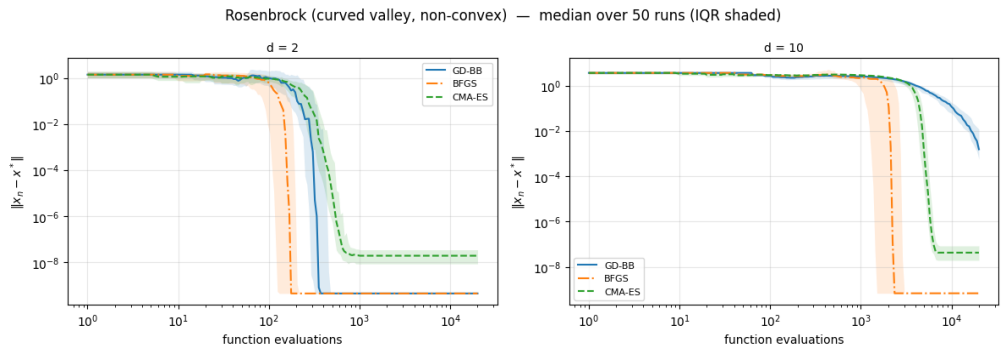
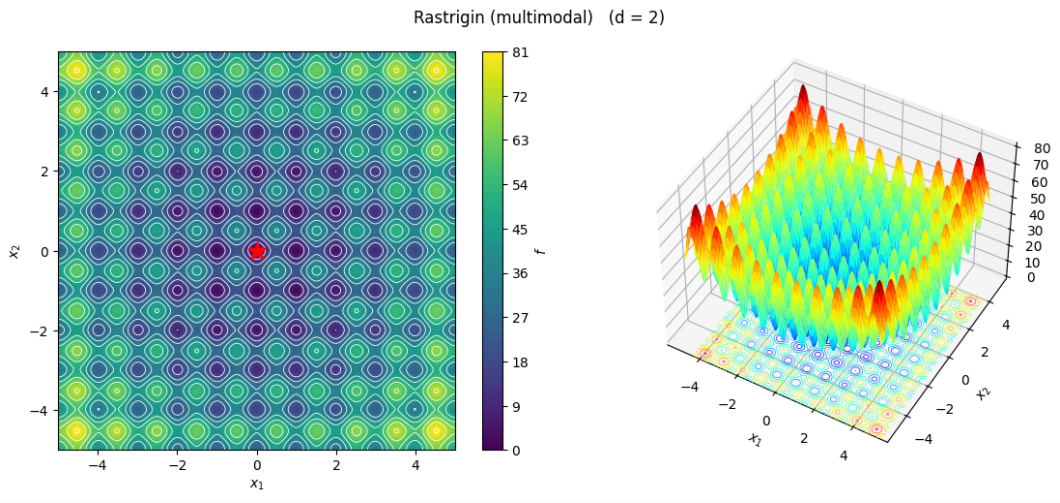


Figure 8: Convergence Analysis on Rosenbrock Problem

Rastrigin Problem



Convergence Analysis: $\|\mathbf{x}_n - \mathbf{x}^*\|$

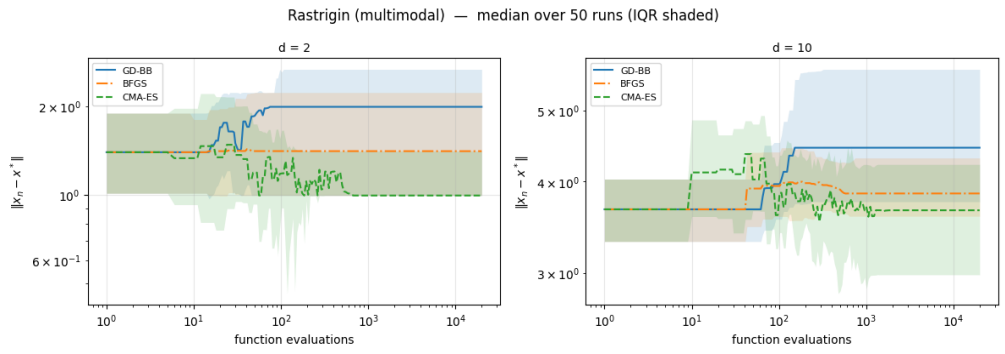


Figure 9: Convergence Analysis on Rastrigin Problem

Success Rate: Runs Reaching $\|\mathbf{x}_n - \mathbf{x}^*\| < 10^{-6}$

Table: Percentage of the 50 runs reaching the target accuracy 10^{-6} within the budget of 2×10^4 function evaluations. Ill-conditioning and rotation break GD at $d = 10$; only Rastrigin resists all three methods.

	$d = 2$			$d = 10$		
Function	GD	BFGS	CMA-ES	GD	BFGS	CMA-ES
Sphere	100%	100%	100%	100%	100%	100%
Ellipsoid	100%	100%	100%	0%	100%	100%
Rotated ellipsoid	100%	100%	100%	0%	100%	100%
Rosenbrock	100%	100%	100%	0%	84%	**88%**
Rastrigin	2%	12%	**24%**	0%	0%	0%

At $d = 2$ every method solves every landscape except Rastrigin. Dimension is what separates them: GD fails on all four hard landscapes at $d = 10$, while CMA-ES matches BFGS everywhere and is slightly ahead on Rosenbrock.

Overall Evaluation

When Gradient Descent is Better

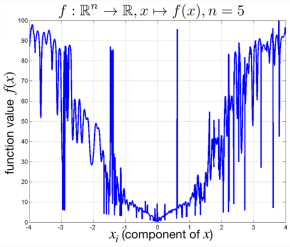
- **Isotropic & well-conditioned problems** (e.g., Sphere).
- **Execution Speed:** Extremely fast due to exact analytical gradient steps.
- Highly scalable to large dimensions **if** the landscape is smooth and convex.

When CMA-ES is Better

- **Ill-conditioned & correlated landscapes** (e.g., Rotated Ellipsoid): Effectively adapts its covariance matrix to match the problem's topography.
- **Non-convex & multimodal problems** (e.g., Rosenbrock, Rastrigin): Its stochastic nature helps escape narrow valleys and shallow local minima.
- Does not require gradient information (derivative-free optimization).

Gradient-based Opt.

Derivative-Free Opt.



Gradient Methods ← gradient/derivative

Random Search ← generate random points

(μ, λ)-ES ← random search + repeatition

CMA-ES ← learning/adaptation (non-trivial

Presented Methods

+Convergence and Comparison

References

Nikolaus Hansen. How to evolve gradient descent into evolution strategies and CMA-ES. Presentation slides, Symposium on Evolutionary Algorithms: Back to the Future and Beyond, Delft, Netherlands, September 2019. URL `http://www.cmap.polytechnique.fr/~nikolaus.hansen/cmaes-from-gradient-descent-delft2019.pdf`.

R.C. White. *A survey of random methods for parameter optimization*. EUT report. E, Fac. of Electrical Engineering. Technische Hogeschool Eindhoven, 1970.

Granino Arthur Korn. *Random-Process Simulation and Measurements*. McGraw-Hill, New York, 1966.

Nahum Aguirre, Erik Cuevas, Alberto Luque-Chang, Oscar Barba-Toscano, and Mario Vasquez-Franco. Covariance matrix adaptation evolution strategy (CMA-ES): A comprehensive survey of variants and hybridizations. *Archives of Computational Methods in Engineering*, 2026. doi: 10.1007/s11831-026-10557-z.

Thank you for your attentions!!! :-)